

SURYANSH KUMAR PATHAK

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EDUCATION

Shri Ramswaroop Memorial University

B.Tech, Computer Science & Engineering — CGPA: 7.00

Lucknow, India

2023 – 2027

M.V.V.M. School, Ballia

2019 – 2021

C.B Inter College, Ballia

2021 – 2023

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy, Scikit-learn, TensorFlow/Keras, LangChain, RAG), C++

Data Analysis & Reporting: Data Cleaning, Exploratory Data Analysis

Data Visualization: Matplotlib, Seaborn

Databases: SQL

AI & Machine Learning: Model Prediction, Classification, Regression, LLM (Fine-tuning), LoRA & QLoRA, NLP, LSTM, Transformer, Generative AI

Community & Tools: Git, GitHub, Hugging Face, Kaggle

Tools & Platforms: Visual Studio Code, Google Colab, Jupyter Notebook

PROJECTS

IMDB Sentiment Analysis Web Application

- Developed an LSTM-based sentiment classifier using pretrained GloVe embeddings, achieving ~88% accuracy on IMDB reviews
- Built an interactive Streamlit web app supporting batch review analysis with real-time predictions
- Generated IMDB-style ratings and sentiment distribution visualizations using Matplotlib
- Implemented robust preprocessing pipeline and ensured consistency between training and deployment
- Containerized the application with Docker for production-ready deployment

Tools: Python, TensorFlow/Keras, LSTM, GloVe, NLP, Streamlit, Docker, Pandas, NumPy

RAG Copilot – AI Document Question Answering System

- Built an AI application that answers questions from uploaded PDF documents
- Developed a document summarization feature to generate short and detailed summaries
- Used LangChain, ChromaDB, and Google Gemini to improve question-answering accuracy
- Implemented hybrid search and re-ranking to retrieve the most relevant document content

Tools: Python, Streamlit, LangChain, ChromaDB, Hugging Face, Google Gemini, Git

AI Medical Assistant

- Developed an intelligent AI medical assistant capable of analyzing symptoms, answering healthcare-related queries, generating preliminary disease predictions, and providing medicine/recommendation support using Large Language Models (LLMs) and machine learning techniques
- Integrated FastAPI backend, vector database, and RAG pipeline
- Containerized application using Docker and deployed scalable services

Tools: Python, Transformers, LangChain, FastAPI, Docker

ACHIEVEMENTS

- Smart India Hackathon: Selected among top teams in college-level Smart India Hackathon (SIH) for presenting an innovative solution to a real-world problem
- Completed 250+ LeetCode DSA Challenges, demonstrating strong problem-solving skills in data structures and algorithms
- Certified by L&T for Data Science Specialist